Early atmosphere. Earth formed about 4.5 billion years ago. Its first atmosphere was mostly hydrogen and helium, which escaped into space because Earth's gravity was too weak to hold them.

Second atmosphere. Volcanic eruptions released water vapor, carbon dioxide, and nitrogen. As the planet cooled, the water vapor condensed and formed the oceans.

Oxygen. About 2.4 billion years ago, cyanobacteria began producing oxygen through photosynthesis. This Great Oxidation Event slowly raised oxygen levels and eventually made complex life possible.